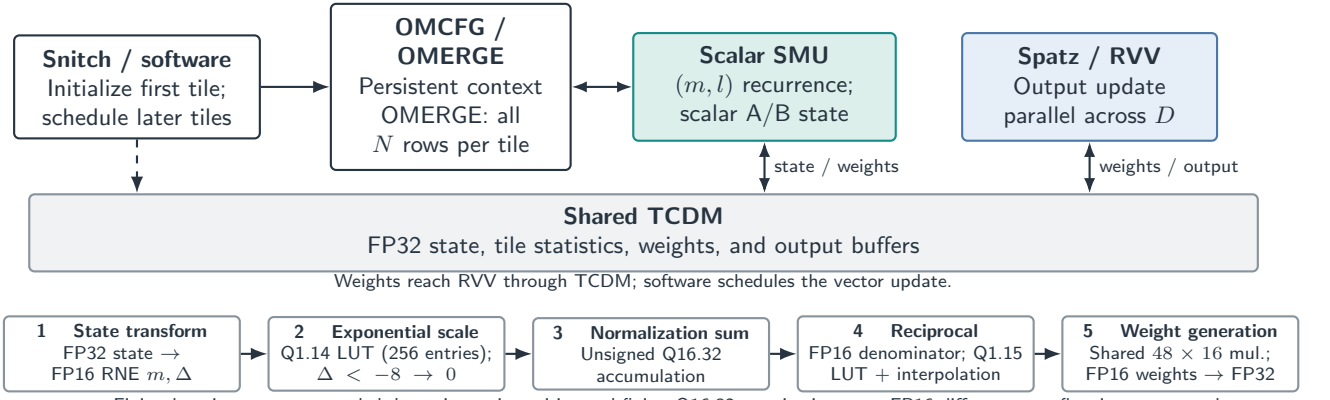

Snitch / software
Initialize first tile; schedule later tiles
OMCFG / OMERGE
Persistent context
OMERGE: all N rows per tile
Scalar SMU
(m, l) recurrence; scalar A/B state
Spatz / RVV
Output update parallel across D
state / weights
weights / output
Shared TCDM
FP32 state, tile statistics, weights, and output buffers
Weights reach RVV through TCDM; software schedules the vector update.
1 State transform
FP32 state → FP16 RNE m, Δ
2 Exponential scale
Q1.14 LUT (256 entries); $\Delta < -8 \rightarrow 0$
3 Normalization sum
Unsigned Q16.32 accumulation
4 Reciprocal
FP16 denominator; Q1.15 LUT + interpolation
5 Weight generation
Shared 48×16 mul.; FP16 weights → FP32
Finite-domain contract: rounded denominator is positive and finite; Q16.32 remains in range; FP16 difference overflow is unsupported.
Numerical stages are conceptual; this strip does not imply pipeline depth or overlap.